Answer **one** question from this Section in the space provided.

7 **(a)** State and define the unit for magnetic flux.

.....

.....

..... [2]

(b) The plan view (from top down) of a train braking system is illustrated in Fig 7.1. The train carriage of mass *m* is mounted on a rectangular metal frame ABCD of length *L* and width *w*, the effective resistance of the frame is *R.* The train carriage is initially moving at a constant speed along the rails.

A uniform magnetic field *B* is directed perpendicularly into the ground over a rectangular region of length *L*. Line P denotes the start of this region while line Q denotes the end of the region. After passing through the magnetic field, the train speed is expected to be reduced to a very low speed after which brakes can be applied to stop it completely. You may assume that friction is negligible.

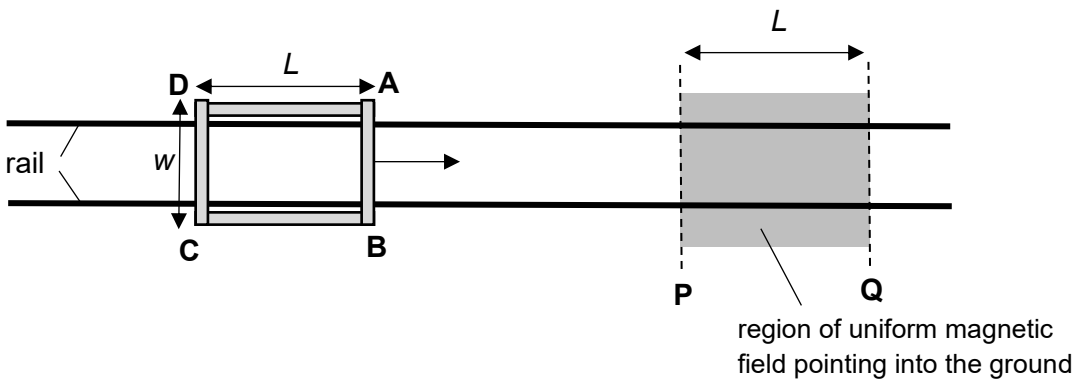

Fig 7.1 (top view)

(i) Explain how the train carriage is slowed as AB moves through the magnetic field from P to Q.

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.....

.....

..... [3]

(ii) Show that the emf induced in the frame *E* is given by *Bwv* where *v* is the speed of the train carriage. Explain your working clearly. [2]

(iii) Hence, deduce an expression for the magnitude of the acceleration of the train carriage as it moves through the magnetic field in terms of *B*, *w*, *v*, *m* and *R*.

acceleration = …….. [3]

(c) The graph in Fig 7.2 shows the velocity of the train carriage as it moves through the magnetic field, from the instant AB crosses line P to the instant CD crosses line Q.

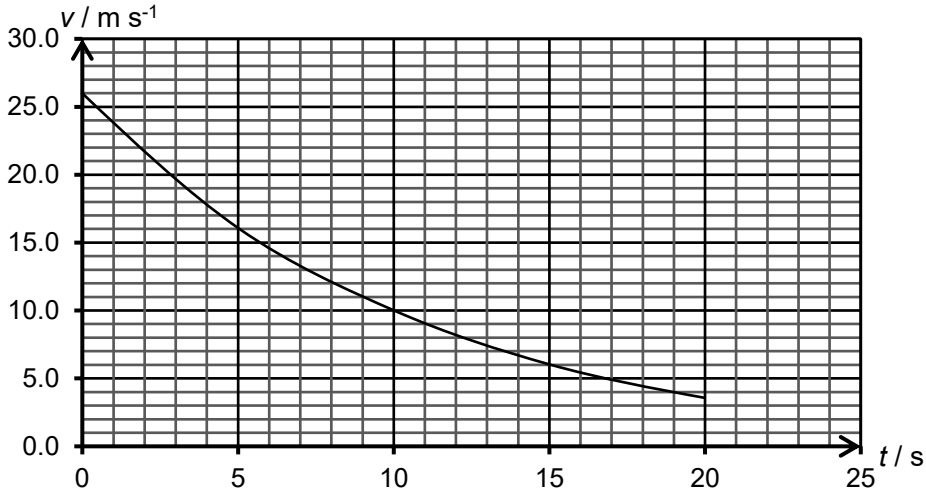

Fig 7.2

(i) Use Fig 7.2 to estimate the distance PQ.

PQ = ... m [3]

(ii) Comment on how increasing the region of magnetic field (distance between P and Q) would affect the exit speed of the train after passing through it.

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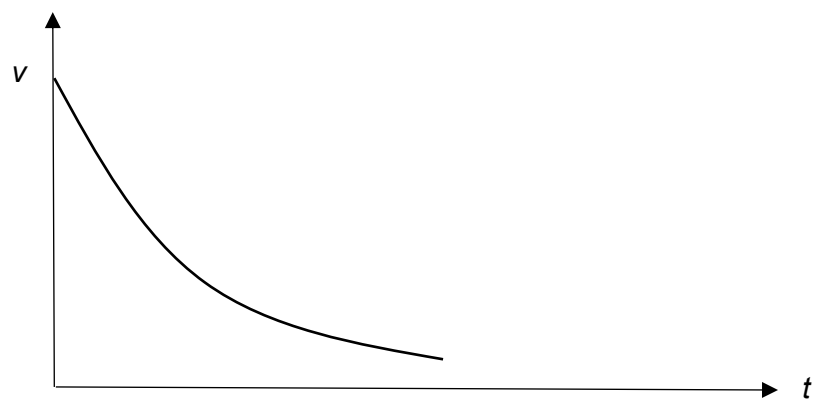
..... [2]

(iii) Sketch on Fig 7.3 the variation of the velocity of the train carriage as it passes through the magnetic field if distance PQ is now reduced.



[3]

Fig 7.3

(iv) Suggest and explain one modification that can be incorporated into the train braking system so that in the event of a train malfunction which causes the train to be moving at a speed that is much higher than expected, the system can still slow the train down to an acceptable speed.

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..... [2]